

through VR6 will be done to get 1.2V, 2V, 3V, 4V, 5V and VR6 for 5.5V, respectively.

The power supply has three output connectors, namely, CON2, CON3 and CON4, connected in parallel. These can be used for connecting loads, voltmeters, oscilloscopes or any other equipment.

Construction and testing

An actual-size, single-side PCB for the power supply is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box such that fixed and variable outputs can be easily taken from the front panel. Ensure proper wiring to avoid any mistake.

Fig. 3: Component layout

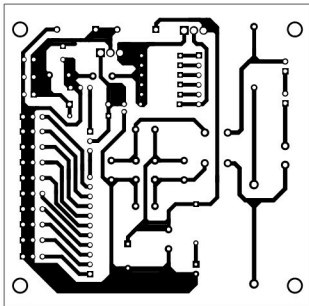


Fig. 2: Actual-size PCB of the power supply

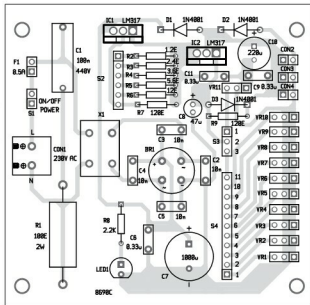


Fig. 3: Component layout of the PCB

reducing the output current. That will ensure protection of regulators.

Output pin of LM317T is connected with the metallic part of the

PARTS LIST

Semiconductors:

IC1, IC2	- LM317 adjustable voltage regulator
BR1	- 1A bridge rectifier
D1-D3	- 1N4001 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1	- 100-ohm, 2W
R2	- 1.2-ohm
R3	- 2.4-ohm
R4	- 3.6-ohm
R5	- 5.6-ohm
R6	- 12-ohm
R7, R9	- 120-ohm
R8	- 2.2-kilo-ohm
VR1-VR11	- 1.5-kilo-ohm potmeter

Capacitors:

C1	- 100nF, 400V
C2-C5	- 10nF ceramic disk
C6, C9, C11	- 0.33μF ceramic disk
C7	- 1000μF, 35V electrolytic
C8	- 47μF, 35V electrolytic
C10	- 220μF, 35V electrolytic

Miscellaneous:

CON1	- 2-pin connector terminal
CON2-CON4	- 2-pin connector
S1	- On/off switch
S2	- One-pole 5-way rotary switch
S3	- One-pole 2-way rotary switch
S4	- One-pole 10-way rotary switch
X1	- 230V AC primary to 22V AC, 1.5A secondary transformer
F1	- 0.5A fuse with holder - Heat sink for LM317 - 230V AC, 50Hz mains power supply

package and, consequently, the metallic part of at least one of the ICs should be isolated from the common heat sink. ●

Petre Tzv Petrov was a researcher and assistant professor at Technical University of Sofia, Bulgaria, and expert-lecturer at OFPPT, Casablanca, Kingdom of Morocco. He is currently working as an electronics engineer in the private sector in Bulgaria



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